

Verification Report (PA#0–PA#20)

Scope

- Audited files: `crypto/*.py` , `api/server.py` , `frontend/src/*` , `web/*` , `README.md` .
- Method: static requirement-to-code verification against `pois_project_full.txt` .
- Runtime/tooling note: command-based validation is currently blocked in this environment because `pwsh.exe` is unavailable.

Overall verdict

The repository is **not fully implemented/correct end-to-end** for all required PAs.

Result	Count	PAs
✅ Pass (static)	10	2, 4, 7, 8, 10, 11, 16, 17, 18, 19
⚠️ Partial	9	0, 1, 3, 5, 9, 13, 14, 15, 20
❌ Fail	2	6, 12

Critical blockers

1. PA#13 `gen_prime` interface mismatch breaks PA#12 keygen path

Requirement expects `gen_prime(bits) -> int` (`pois_project_full.txt:2136-2137`), but implementation returns tuple (prime, attempts) (`crypto/pa13_miller_rabin.py:165-176`), while PA#12 uses it as an int (`crypto/pa12_rsa.py:58-65`).

2. PA demo API uses nonexistent request field `req.foundation` for PA#3/5/6 demos

`DemoRequest` has only `pa` and `params` (`api/server.py:32-35`), but PA demo handlers access `req.foundation` (`api/server.py:372` , `394` , `404`).

3. PA#6 demo wiring mismatch

API constructs `CCAEncryption(prf=...)` (`api/server.py:405`) but constructor is `CCAEncryption(enc=None, mac=None)` (`crypto/pa06_cca_enc.py:33-40`).

PA-by-PA verification matrix

PA	Status	Verification note	Evidence
0	 Partial	UI scaffold exists, but several required PA#0 behaviors are incomplete: no explicit foundation modules (AESFoundation / DLPFoundation), no real-time recompute-on-input-change, no explicit Forward/Backward mode toggle, no shortest-path routing implementation, and reduce path is direct-lookup only.	Requirements: pois_project_full.txt:710-733 , 725-731 ; implementation: api/server.py:43-62 , 210-214 ; UI action buttons: frontend/src/components/Explorer.jsx:104-151 ; no foundation modules found.
1	 Partial	Core OWF/PRG logic and tests are present, but required seed(s), next_bits(n) interface is overwritten by duplicate method definitions (non-stateful signatures override stateful ones).	Requirement: pois_project_full.txt:859 ; duplicate defs: crypto/pa01_owf_prg.py:200-223 vs 252-258 , and 279-299 vs 343-353 .
2	 Pass (static)	GGM PRF, AES PRF plug-in, PRF⇒PRG, and distinguishing game are implemented with expected interfaces.	crypto/pa02_prf_ggm.py:31-81 , 151-173 , 245-255 , 287-320 .
3	 Partial	Core CPA scheme and IND-CPA game exist, but PA demo/API integration is broken due req.foundation access on DemoRequest .	Core: crypto/pa03_cpa_enc.py:50-112 , 140-179 ; API issue: api/server.py:32-35 , 372 .
4	 Pass (static)	CBC/OFB/CTR modes, unified API, and reuse-attack demos exist.	crypto/pa04_modes.py:177-228 , 235 , 265 .
5	 Partial	PRF-MAC, CBC-MAC, EUF-CMA, and HMAC stub are	Requirement: pois_project_full.txt:1300-1301 ; present

PA	Status	Verification note	Evidence
		present, but PA#5-required naive single-hash-MAC length-extension demonstration is missing in this module.	features: <code>crypto/pa05_mac.py:106-114</code> , 219-261 ; no corresponding length-extension demo in PA5 implementation.
6	 Fail	Core EtM class exists, but repo-level PA6 path is broken in API/demo integration (<code>req.foundation</code> bug + wrong constructor arg).	<code>api/server.py:32-35</code> , <code>404-405</code> ; <code>crypto/pa06_cca_enc.py:33-40</code> .
7	 Pass (static)	Generic Merkle-Damgård framework, MD-strengthening padding, trace support, and collision-propagation demo are implemented.	<code>crypto/pa07_merkle_damgard.py:21-93</code> , <code>95-139</code> , 177-216 .
8	 Pass (static)	DLP group setup, DLP compression, Merkle-Damgård integration, and collision demo are implemented.	<code>crypto/pa08_dlp_crhf.py:26-50</code> , <code>57-93</code> , 103-138 , <code>161-188</code> .
9	 Partial	Naive attack is implemented, but Floyd attack does not return collision pair; truncated DLP attack report does not include the two colliding inputs as required.	Requirement: <code>pois_project_full.txt:1683-1684</code> , <code>1696-1699</code> ; implementation: <code>crypto/pa09_birthday.py:131-136</code> , <code>217-224</code> .
10	 Pass (static)	HMAC, constant-time verify, CRHF $\leftrightarrow$ MAC demos, length-extension demo, and Encrypt-then-HMAC primitives are implemented.	<code>crypto/pa10_hmac.py:31-96</code> , <code>102-113</code> , <code>194-250</code> , 257-286 , <code>292-362</code> .
11	 Pass (static)	DH key exchange, MITM demo, and CDH hardness demo are implemented.	<code>crypto/pa11_diffie_hellman.py:22-83</code> , <code>90-128</code> , 135-169 .
12	 Fail	RSA keygen path is broken by PA#13 return-type mismatch,	Requirement: <code>pois_project_full.txt:2076-2078</code> ; keygen call:

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		and Bleichenbacher task is only oracle illustration (not adaptive plaintext recovery implementation).	crypto/pa12_rsa.py:58-65 ; oracle-only demo: crypto/pa12_rsa.py:255-293 .
13	 Partial	Miller-Rabin core is implemented, but required interface is mismatched (gen_prime(bits) -> int expected, tuple returned), and CLI/demo section contains inconsistent keys.	Requirement: pois_project_full.txt:2136-2137 ; impl: crypto/pa13_miller_rabin.py:165-176 ; inconsistent CLI keys: 309-310 .
14	 Partial	CRT, CRT-decrypt benchmark, Håstad attack, and padding defense demo are present; explicit “attack boundary” computation/report is not implemented.	Requirement: pois_project_full.txt:2315-2317 ; implementation coverage: crypto/pa14_crt.py:23-109 , 154-214 , 217-243 .
15	 Partial	Signature scheme and attack demos are present; interface wrapper deviates from requested explicit Sign(sk,m) / Verify(vk,m,sigma) semantics by expecting a DigitalSignature object directly.	Requirement: pois_project_full.txt:2385 ; wrapper: crypto/pa15_signatures.py:178-184 .
16	 Pass (static)	ElGamal keygen/enc/dec, malleability attack, IND-CPA game, and tiny-group distinguisher are implemented.	crypto/pa16_elgamal.py:42-63 , 69-118 , 146-190 , 196-276 .
17	 Pass (static)	Encrypt-then-sign (signcrypt style), verify-then-decrypt ordering, CCA2 game, and contrast demo are implemented.	crypto/pa17_cca_pkc.py:51-87 , 93-133 , 139-176 .

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18	 Pass (static)	3-step OT API, correctness tests, and sender/receiver privacy demos are implemented.	<code>crypto/pa18_ot.py:50-140</code> , <code>190-252</code> , <code>254-277</code> .
19	 Pass (static)	Secure AND/XOR/NOT primitives, truth-table testing, and privacy reasoning are implemented.	<code>crypto/pa19_secure_and.py:48-142</code> , <code>149-196</code> , <code>203-244</code> .
20	 Partial	Circuit + secure evaluation + mandatory circuits + lineage/performance are present, but required transcript-simulatability privacy verification for the three circuits is not implemented.	Requirement: <code>pois_project_full.txt:2755-2758</code> ; implementation has no corresponding privacy verification routine (only header mention at <code>crypto/pa20_mpc.py:11</code>).

Additional repo-level observations

- `README.md` claims full completion (`README.md:5`), but current codebase has the blockers above.
- “No external crypto library” condition appears respected in crypto modules (custom implementations under `crypto/`).